Program - 5

Aim:- Develop DFD model (level-0, level-1 DFD & Data dictionary) of the project.

Overall discription:-

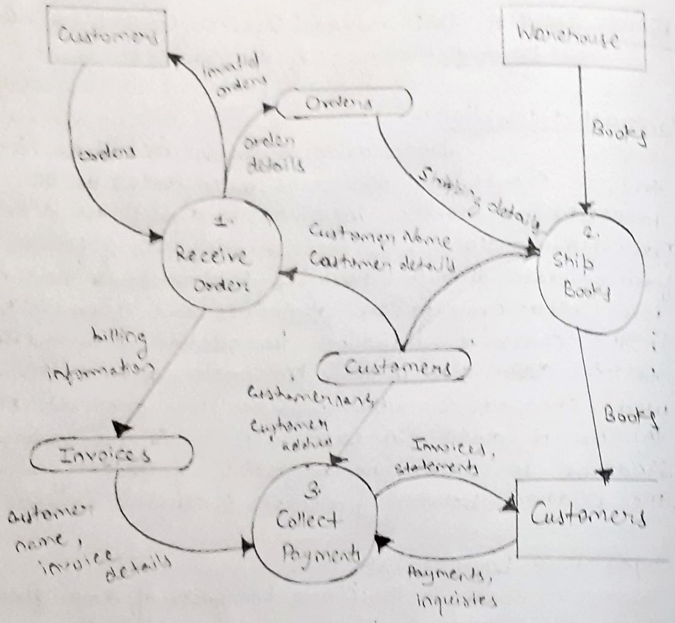
Data drive business activities & can trigger events (e.g. new sales order data) or be processed to provide information about the activity. Data flow analysis, as the name suggests, follows the flow of data through business processes & determine how ~~reco.~~ organisation objectives are accomplished. In the course of handling transaction & completing tasks, data are input, processed, stored, retrived, used, changed & output. Data flow analysis studies the use of data in each activity & documents the findings in data flow diagrams, graphically showing the relation between processes & data.

Physical & Logical DFDS:

There are two types of data flow diagrams, namely physical data flow diagrams & logical data flow diagrams & it is important to distinguish clearly between the two:

Physical Data Flow Diagrams:-

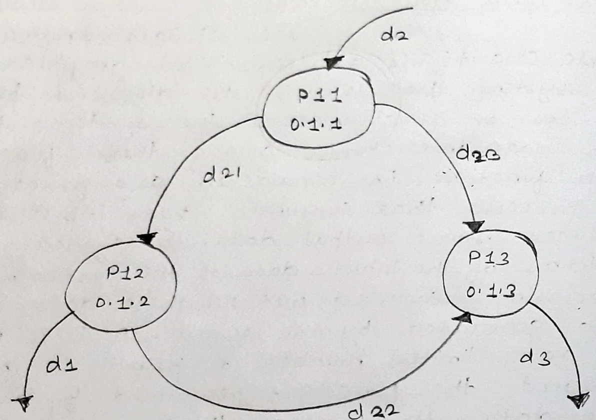
An implementation-dependent view of the current system, showing that tasks are carried out & how they are performed. Physical



characteristics can include:
Name of people
Form and document names or numbers
Master and transaction files
Equipment & devices used

Logical Data Flow Diagrams:-
The DFD (also known as a bubble chart) is a hierarchical graphical model of a system that shows the different processing activities or functions that the systems perform & the data interchange among these functions. Each function is considered as a processing station (or process) that consumes some input data & produces some output data. The system is represented in terms of the input data of the system, various processing carried out on these data, & the output data generated by the system. A DFD model uses a very limited number of primitive symbols to represent the functions performed by a system & the data flow among these functions.

Symbols used for designing DFDs:-
Here, two examples of data flow that describe input & validation of data are considered. The two processes are directly connected by a data flow. This means that the 'validate number' process can start only after the 'read-number' process had supplied data to it. However the two processes are



connected through adata store. Hence, the operations of the two bubbles are independent. The first one is termed 'synchronous' & the second one 'asynchrona

<u>Importance of DFDs in a good software design:-</u>

The main reason why the DFD technique is so popular is probably because of the fact that DFD is a very simple formalism – it is simple to understand & use. Starting with a set of High-level functions that a system performs, a DFD model hierarchically represents various sub-factions. In fact, any hierarchical model is a simple to understand. Human mind is such that it can easily understand any hierarchical model of a system - because in a hierarchical model, starting with a very simple & abstract model of a system, different details of the system are slowely introduced through different hierarchies. The data flow diagramming technique also follows a very simple set of intuitive concepts & rules. DFD is an elegant modeling technique that turns out to be useful not only to represent the results of structured analysis of a software problem, but also for several other applications such as showing the flow of documents or items in an organizations.

Balancing a DFD :- The data that flow into or out of a bubble must match the data flow at the ~~fac~~ next level of DFD. This is known as balancing a DFD. In the level of the DFD, data items d1 and d3 flow out of the bubble 0.1 & the data item d2 flows into the bubble 0.1. In the next level, bubble 0.1 is decomposed. The decomposition is balanced, as d1 & d3 flow out of the level 2 diagram & d2 flow in.

Viva Question :-

Q1 What is DFD ?

Ans A data flow diagram (DFD) is a graphical representation of the 'flow' of data through an information system, modelling its process aspects. A DFD is often used as a preliminary step to create an overview of the system without going into great details, which can later be elaborated.

Q2 What is Level-0 DFD ?

Ans Highest abstraction level DFD is known as level-0 DFD also called a context level DFD, which depicts the entire information system as one diagram concealing all the underlying details.

Q3 Briefly Define Top-Down & Bottom-up Design model.

Ans Top-down model starts with generalized view of System and decomposes it to more specific ones, whereas bottom-up model starts with most specific & basic components first & keeps composing the components to get higher level of abstraction.

Q4 What is a level 2 data flow diagram?

Ans A level 2 data flow diagram (DFD) offers a more detailed look at the processes that make up an information system than a level 1 DFD does. It can be used to plan or record the specific makeup of a system. You can then input the particular of your own system.

Q5 How many level in the DFD? What is Data dictionary?

Ans There are three level in DFD
(i) 0 Level
(ii) 1 Level
(iii) 2 Level

Data dictionary is reffered to as meta-data. Meaning, it is a repository of data about data. Data dictionary is used to organize the names & their references used in system such as objects & files along with their naming conventions.